turn on the seatbelt indicator for the following reasons:

- **Seatbelt Reminder Function** - Each time the cluster receives a battery current input on the fused ignition switch output (run-start) circuit, the indicator will be illuminated as a seatbelt reminder for about seven seconds, or until the ignition switch is turned to the Off position, whichever occurs first. This reminder function will occur regardless of the status of the seatbelt switch input to the cluster.
- **Driver Side Front Seatbelt Not Buckled** - Following the seatbelt reminder function, each time the cluster detects a ground on the seat belt switch sense circuit (seatbelt switch closed = seatbelt unbuckled) with the ignition switch in the Start or On positions, the indicator will be illuminated. The seatbelt indicator remains illuminated until the seat belt switch sense input to the cluster is an open circuit (seatbelt switch open = seatbelt buckled), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the seatbelt indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The seatbelt switch is connected in series between ground and the seat belt switch sense input to the instrument cluster. The seatbelt switch input to the instrument cluster circuitry can be diagnosed using conventional diagnostic tools and methods. For further diagnosis of the seatbelt indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).

SHIFT INDICATOR (TRANSFER CASE)

DESCRIPTION

Fig. 23 4WD Indicator

A four-wheel drive indicator is standard equipment on all instrument clusters (Fig. 23). The four-wheel drive indicator is located near the upper edge of the instrument cluster, to the right of center. The four-wheel drive indicator consists of a stencil-like cutout of the text “4WD” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the “4WD” text to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The illumination intensity of the four-wheel drive indicator is one-step dimmable. When the exterior lighting is turned On, the indicator is dimmed; and, when the exterior lighting is turned Off, the indicator is illuminated at full intensity. The four-wheel drive indicator is serviced as a unit with the instrument cluster.

OPERATION

The four-wheel drive indicator gives an indication to the vehicle operator that a four-wheel drive operating mode of the four-wheel drive transfer case is selected. The four-wheel drive indicator lights when the transfer case is engaged in the 4H or 4L positions. This indicator is controlled by a transistor on the instrument cluster electronic circuit board based upon cluster programming. On models not equipped with the optional Off-Road Package the cluster also uses a hard wired transfer case switch input. Models that are equipped with the Off-Road Package do not have a hard wired transfer case switch input, so the cluster uses electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus to determine the transfer case operating mode. The four-wheel drive indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the four-wheel drive indicator for the following reasons:

- **Transfer Case Switch Input** - Each time the cluster detects a ground on the 4WD switch sense circuit (transfer case switch closed = 4WD mode selected) the part time indicator is illuminated. The indicator remains illuminated until the 4WD switch sense input to the cluster is an open circuit (transfer case switch open = 4WD mode not selected), or until the ignition switch is turned to the Off position, whichever occurs first.
- **4WD Transfer Case Status Message** - Each time the cluster receives a message from the PCM indicating that the transfer case is operating in a four-wheel drive mode, the 4WD indicator is illuminated. The 4WD indicator remains illuminated until the cluster receives messages from the PCM indicating that the transfer case is not in a four-wheel drive mode or until the ignition switch is turned to the Off position, whichever occurs first.